# 📐 Geometry Handbook — Formulas & Figures (SSC CGL)

> Every shape & theorem with a **labelled figure** + its formulas. Companion to
>
> SSC-CGL-ASO-Formulas-Tricks-Master.md.
>
> Syllabus: Quant — Geometry & Mensuration (Para 13.10.3 / 13.11.1.4–13.11.1.5).
>
> View in a monospace font (or the HTML/PDF) so the figures stay aligned.

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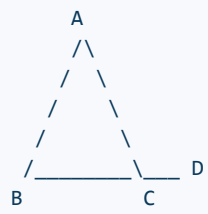

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## 1. Lines & Angles

```
Types of angles:                 Angle pairs on a transversal:
• Acute    < 90°                       a\ b
• Right    = 90°   ┐                     \
• Obtuse   90–180°                  ─────\──────  line 1
• Straight = 180°  ──                  c    \ d
• Reflex   180–360°                         \
                                         e\ f  \
                                    ──────\─────\──  line 2
                                            g\ h
```

| Concept | Rule |
|---|---|
| Complementary angles | sum = **90°** |
| Supplementary angles | sum = **180°** |
| Angles on a straight line | sum = **180°** (linear pair) |
| Angles around a point | sum = **360°** |
| Vertically opposite angles | **equal** (a = d, b = c) |
| Parallel lines: corresponding | **equal** |
| Parallel lines: alternate (interior/exterior) | **equal** |
| Parallel lines: co-interior (allied) | sum = **180°** |

## 2. Triangle — basics & types




- Angle sum ∠A + ∠B + ∠C = 180°
- Exterior angle = sum of 2 remote interiors
  ∠ACD = ∠A + ∠B
- Sum of any two sides > third side
- Side opposite the bigger angle is longer

| By sides | By angles |
|---|---|
| **Equilateral** — all sides equal (each angle 60°) | **Acute** — all angles < 90° |
| **Isosceles** — two sides equal (base angles equal) | **Right** — one angle = 90° |
| **Scalene** — all sides different | **Obtuse** — one angle > 90° |

## 3. The Four Centres of a Triangle

| Centre | Where lines meet | Note |
|---|---|---|
| Centroid G | the 3 MEDIANS (vertex → midpoint) | divides each median 2:1 (from vertex) |
| Incentre I | the 3 ANGLE BISECTORS | centre of inscribed circle |
| Circumcentre O | the 3 PERP. BISECTORS of the sides | centre of circumscribed ○; equidistant from vertices |
| Orthocentre H | the 3 ALTITUDES (vertex ⊥ opposite side) | — |

- G, O, H are collinear → the EULER LINE; and OG : GH = 1 : 2.
- In an equilateral triangle all four centres COINCIDE.

```
      A                    Median: A → midpoint of BC.
      /|\                  Centroid G lies on all three, 2:1 from vertex.
     / | \                 r (inradius)  = Area / s
    /  G  \                 R (circumradius) = (a·b·c) / (4·Area)
   /  /|\  \                In a right triangle: R = hypotenuse / 2
  B__/_|_\__C               and the circumcentre is the midpoint of hypotenuse.
```

## 4. Triangle Area Formulas

| Given | Area |
|---|---|
| base & height | **½ × base × height** |
| 3 sides a,b,c (Heron) | **√[s(s−a)(s−b)(s−c)]**, s = (a+b+c)/2 |
| two sides & included angle | **½ · a · b · sin C** |
| equilateral side a | **(√3 / 4) · a²** |
| with inradius r | **r × s** |
| with circumradius R | **(a·b·c) / (4R)** |

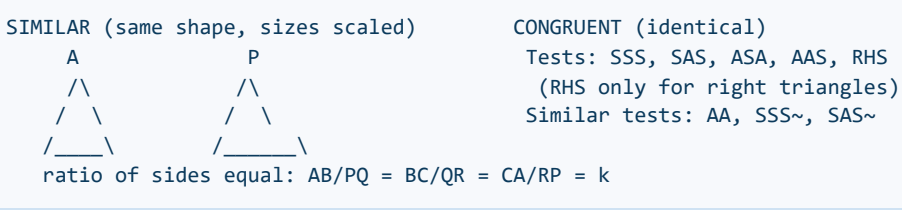
```
Equilateral triangle (side a):
     /\          height  h = (√3/2)·a
    /  \         area      = (√3/4)·a²
   / h  \        inradius  r = a / (2√3)
  /______\       circumradius R = a / √3   (R = 2r)
     a
```

## 5. Pythagoras Theorem & Triples

```
      |\                   c² = a² + b²
      | \                  (hypotenuse² = sum of squares of legs)
    b |  \ c               Common Pythagorean triples:
      |   \                  3, 4, 5      5, 12, 13
      |____\                 8, 15, 17    7, 24, 25
        a                    9, 40, 41    20, 21, 29
                           (and any multiple: 6-8-10, 9-12-15 …)
```

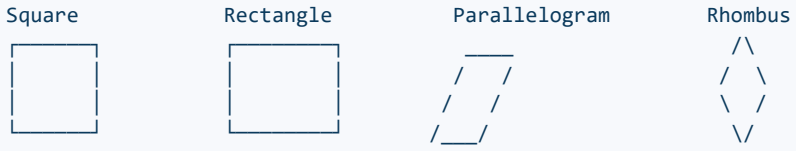
| Result | Formula |
|---|---|
| Hypotenuse | $\sqrt{(a^2 + b^2)}$ |
| A leg | $\sqrt{(c^2 - \text{other leg}^2)}$ |
| Altitude to hypotenuse | (a·b)/c |
| Right-triangle inradius | r = (a + b − c)/2 |

## 6. Similarity & Congruence



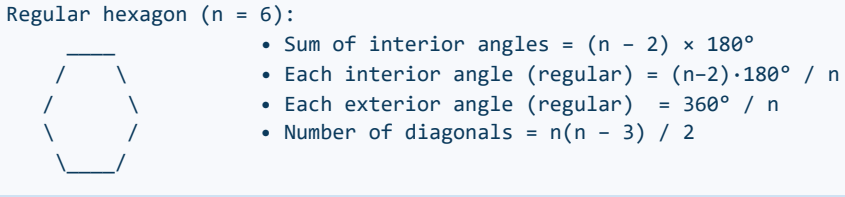

| In similar triangles (ratio k) | Relationship |
|---|---|
| Corresponding sides | in ratio **k** |
| Perimeters | in ratio **k** |
| **Areas** | in ratio **$k^2$** |
| (for solids) Volumes | in ratio **$k^3$** |
| Basic Proportionality (Thales) | a line ∥ one side divides the other two **proportionally** |

## 7. Quadrilaterals



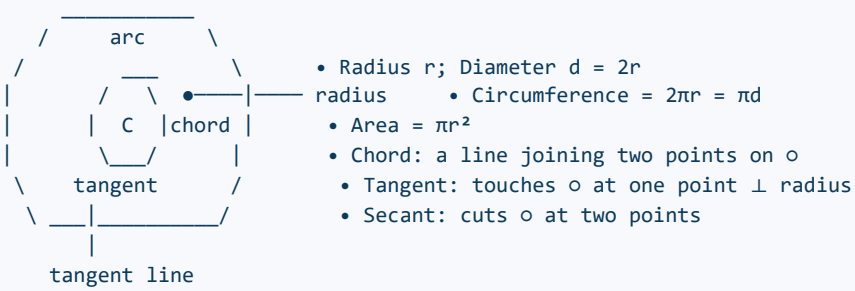

| Shape | Area | Perimeter / notes |
|---|---|---|
| Square (side a) | $a^2$ | 4a; diagonal = $a\sqrt{2}$ |
| Rectangle (l, b) | l·b | $2(l+b)$; diagonal = $\sqrt{(l^2+b^2)}$ |
| Parallelogram | base × height | opposite sides ∥ & equal; diagonals bisect |
| Rhombus (diagonals $d_1$,$d_2$) | **$\frac{1}{2} \cdot d_1 \cdot d_2$** | all sides equal; diagonals ⊥ bisect |
| Trapezium (∥ sides a,b; height h) | **$\frac{1}{2}(a+b) \cdot h$** | one pair of sides parallel |
| Kite (diagonals $d_1$,$d_2$) | $\frac{1}{2} \cdot d_1 \cdot d_2$ | diagonals ⊥ |
| Cyclic quad (Brahmagupta) | $\sqrt{[(s-a)(s-b)(s-c)(s-d)]}$ | opposite angles sum 180° |

## 8. Polygons



| n | Name | Interior-angle sum | Each interior (regular) |
|---|---|---|---|
| 3 | Triangle | 180° | 60° |
| 4 | Quadrilateral | 360° | 90° |
| 5 | Pentagon | 540° | 108° |
| 6 | Hexagon | 720° | 120° |
| 8 | Octagon | 1080° | 135° |

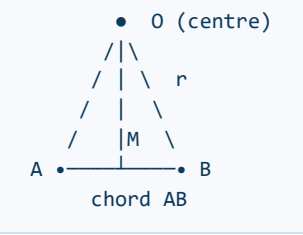
## 9. Circle — parts & theorems




| Theorem | Statement |
|---|---|
| Angle at centre | = **2 ×** angle at circumference (same arc) |
| Angles in same segment | **equal** |
| Angle in a semicircle | **90°** (diameter subtends right angle) |
| Cyclic quadrilateral | opposite angles sum **180°** |
| Tangent ⊥ radius | at the point of contact (90°) |
| Two tangents from a point | are **equal** in length |
| Alternate segment | tangent-chord angle = angle in alternate segment |
| Intersecting chords | PA · PB = PC · PD |
| Equal chords | equidistant from the centre |

| Circle measure | Formula |
|---|---|
| Area | $\pi r^2$ |
| Circumference | $2 \pi r$ |
| Arc length (angle θ°) | $(\theta/360) \times 2\pi r$ |
| Sector area (angle θ°) | $(\theta/360) \times \pi r^2$ |
| Segment area | sector area − triangle area |

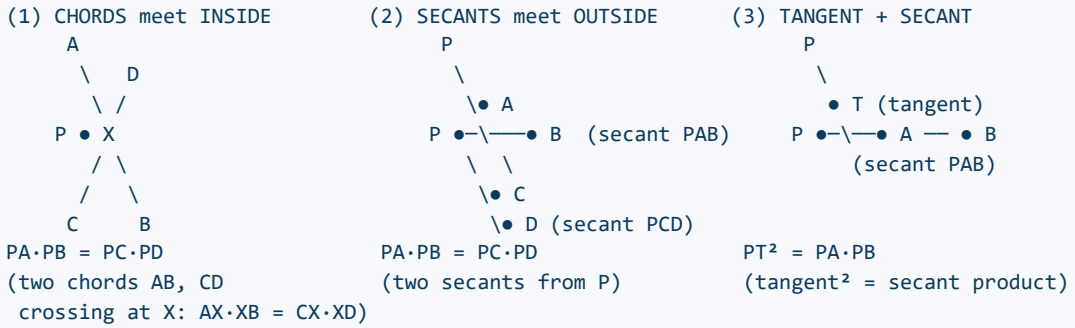
## 9A. Circle — Chord Theorems




- $OM \perp AB \Rightarrow AM = MB$
- Distance of chord from centre: $OM = \sqrt{(r^2 - (AB/2)^2)}$
- Conversely, the line from O to the midpoint of a chord is $\perp$ to it.

| Chord theorem | Statement |
|---|---|
| Perpendicular from centre | **bisects** the chord (AM = MB) |
| Line from centre to chord-midpoint | is **perpendicular** to the chord |
| Equal chords | are **equidistant** from the centre (and vice-versa) |
| Longer chord | is **nearer** the centre; the diameter is the longest chord |
| Equal chords subtend | **equal angles** at the centre |

## 9B. Intersecting Chords, Secants & Tangent–Secant (Power of a Point) ⭐



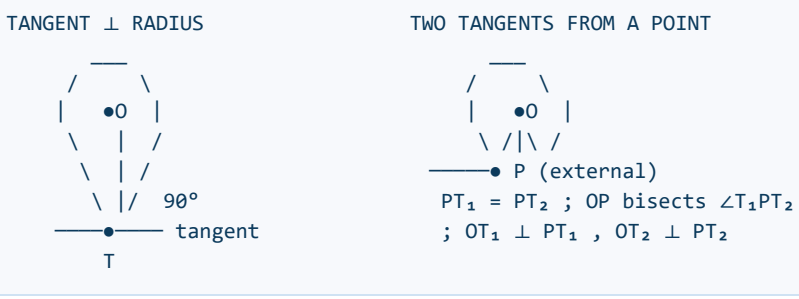

| Theorem (Power of a Point) | Relation |
|---|---|
| Two chords intersect **inside** at X | $AX \cdot XB = CX \cdot XD$ |
| Two secants from an **external** point P | $PA \cdot PB = PC \cdot PD$ |
| **Tangent–Secant** from external point P | $PT^2 = PA \cdot PB$ |
| Two **tangents** from external point P | $PT_1 = PT_2$ (equal lengths) |

 **One idea:** for any point P and a line through it cutting the circle at two points,

the **product of the two distances is constant** (the "power" of P). Inside → chord·chord;

outside → secant·secant; tangent is the limiting case where the two points merge → $PT^2$.

## 9C. Tangent Theorems




| Tangent theorem | Statement |
|---|---|
| Tangent ⊥ radius | at the point of contact (90°) |
| Two tangents from external point | are **equal** ($PT_1 = PT_2$) |
| Line from centre to external point | **bisects** the angle between the two tangents |
| Tangent length | $PT = \sqrt{(OP^2 - r^2)}$ |
| Alternate Segment Theorem | angle between tangent & chord = angle in the **alternate segment** |

```
DIRECT & TRANSVERSE COMMON TANGENTS (two circles, centres d apart, radii R, r)
• Length of DIRECT common tangent     = √(d² − (R − r)²)
• Length of TRANSVERSE common tangent = √(d² − (R + r)²)
```

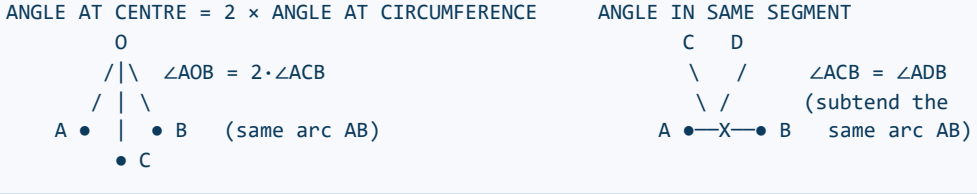
**Number of common tangents by position of two circles** (centres `d` apart, radii `R ≥ r`):

| Condition | Position of Circles | Common Tangents |
|---|---|---|
| `d > R + r` | Separate (do not touch) | 4 |
| `d = R + r` | Externally touching | 3 ⭐ |
| `R − r < d < R + r` | Intersecting | 2 |
| `d = R − r` | Internally touching | 1 ⭐ |
| `d < R − r` | One circle inside another | 0 |
| `d = 0, R = r` | Coincident (same circle) | Infinite |

🔑 *Trick:* Tangent count drops as the circles get closer — the two **touching** cases (⭐) are the exam favourites: externally touching → **3**, internally touching → **1**.

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## 9D. Proportionality & Angle Theorems (Circle)

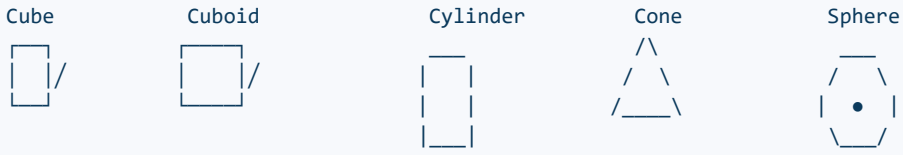


| Angle / proportion theorem | Statement |
|---|---|
| Angle at centre | = **2 ×** angle at the circumference on the same arc |
| Angles in the same segment | are **equal** |
| Angle in a semicircle | = **90°** |
| Cyclic quadrilateral | opposite angles are **supplementary** (sum 180°) |
| Exterior angle of cyclic quad | = interior **opposite** angle |
| Ptolemy's theorem (cyclic quad) | AC · BD = AB · CD + AD · BC (product of diagonals) |
| Equal arcs | subtend equal angles at the centre & equal chords |

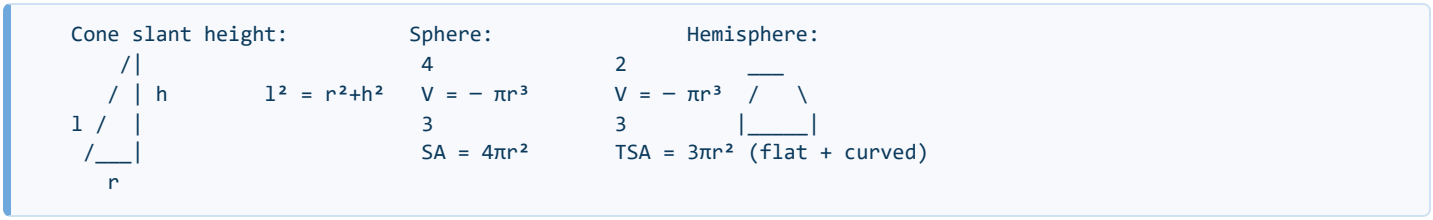
## 10. Mensuration 2D — areas & perimeters

| Shape | Area | Perimeter |
|---|---|---|
| Square (a) | $a^2$ | 4a |
| Rectangle (l,b) | l·b | 2(l+b) |
| Triangle (b,h) | ½·b·h | a+b+c |
| Equilateral (a) | $(\sqrt{3}/4)a^2$ | 3a |
| Parallelogram | b·h | 2(a+b) |
| Rhombus ($d_1$,$d_2$) | ½·$d_1$·$d_2$ | 4·side |
| Trapezium (a,b,h) | ½(a+b)h | sum of sides |
| Circle (r) | $\pi r^2$ | $2\pi r$ |
| Semicircle (r) | $\frac{1}{2}\pi r^2$ | $\pi r + 2r$ |
| Ring (R,r) | $\pi(R^2-r^2)$ | — |

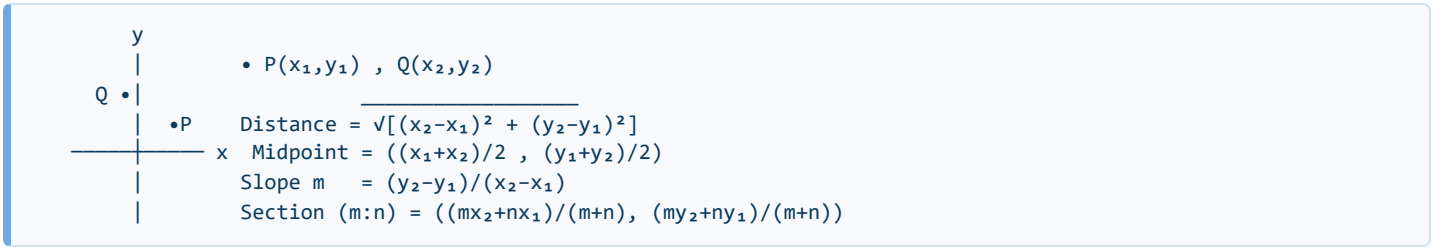
## 11. Mensuration 3D — solids



| Solid | Volume | Curved/Lateral SA | Total SA |
|---|---|---|---|
| Cube (a) | $a^3$ | $4a^2$ | $6a^2$; diagonal $a\sqrt{3}$ |
| Cuboid (l,b,h) | $l \cdot b \cdot h$ | $2h(l+b)$ | $2(lb+bh+hl)$; diag $\sqrt{(l^2+b^2+h^2)}$ |
| Cylinder (r,h) | $\pi r^2 h$ | $2\pi rh$ | $2\pi r(r+h)$ |
| Cone (r,h) | $\frac{1}{3}\pi r^2 h$ | $\pi rl$ | $\pi r(r+l)$; slant $l=\sqrt{(r^2+h^2)}$ |
| Sphere (r) | $4/3 \cdot \pi r^3$ | — | $4\pi r^2$ |
| Hemisphere (r) | $2/3 \cdot \pi r^3$ | $2\pi r^2$ | $3\pi r^2$ |
| Frustum (R,r,h) | $\frac{1}{3}\pi h(R^2+r^2+Rr)$ | $\pi(R+r)l$ | $+ \pi(R^2+r^2)$ |




## 12. Coordinate Geometry




| Concept | Formula |
|---|---|
| Distance | $\sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$ |
| Midpoint | $((x_1+x_2)/2, (y_1+y_2)/2)$ |
| Slope m | $(y_2-y_1)/(x_2-x_1)$ |
| Line (slope-intercept) | $y = mx + c$ |
| Parallel lines | equal slopes ($m_1 = m_2$) |
| Perpendicular lines | $m_1 \cdot m_2 = -1$ |
| Area of triangle (vertices) | $\frac{1}{2}\ \backslash$ | $x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\backslash$ | |

## 13. Quick recall table

| Prompt | Answer |
|---|---|
| Triangle angle sum | 180° |
| Equilateral area | $(\sqrt{3}/4)\cdot a^2$ |
| Heron's area | $\sqrt{[s(s-a)(s-b)(s-c)]}$ |
| Centroid divides median | 2 : 1 |
| Circumradius R | abc/(4·Area) |
| Inradius r | Area/s |
| Angle in semicircle | 90° |
| Angle at centre vs circumference | 2 : 1 |
| Rhombus area | $\frac{1}{2}\cdot d_1\cdot d_2$ |
| Trapezium area | $\frac{1}{2}(a+b)\cdot h$ |
| Interior-angle sum (n-gon) | $(n-2)\cdot 180°$ |
| Diagonals of n-gon | $n(n-3)/2$ |
| Cube diagonal | $a\sqrt{3}$ |
| Cuboid diagonal | $\sqrt{(l^2+b^2+h^2)}$ |
| Cone slant l | $\sqrt{(r^2+h^2)}$ |
| Sphere volume / SA | $4/3\cdot\pi r^3$ / $4\pi r^2$ |
| Hemisphere TSA | $3\pi r^2$ |
| Areas of similar figures | ratio of sides$^2$ |
| Perpendicular slopes | $m_1\cdot m_2 = -1$ |

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✅ **Strategy:** learn the **figure first**, then attach its formulas — recall is far

stronger when you can *picture* the shape. Redraw 2–3 figures from memory daily.

Back to the main formula sheet ».